**Tool Analysis**

- Lucidchart: While the free tier offers a capable node-based canvas for flowcharts and PERT diagrams, the Gantt functionality is limited. The platform requires manual formatting for timelines and lacks automatic calculation for task dependencies.
- TeamGantt: This specialized tool features a drag-and-drop interface that automates duration and overlap calculations. However, it does not support the custom node-based architecture required for PERT charts.
- Draw.io (diagrams.net): Although it is an open-source, cost-effective solution for custom PERT node shapes, it offers no timeline automation, making Gantt chart creation a purely manual and inefficient process.
- Microsoft Project: As the industry standard, it handles both PERT and Gantt generation concurrently. However, the licensing costs and steep learning curve make it impractical for this project.

**Selection & Justification**

The group opted to use Lucidchart for the PERT chart and TeamGantt for the Gantt chart.

We selected Lucidchart for the PERT requirements because its interface allows us to instantiate custom node shapes to display specific variables, task name, expected time, Earliest/Latest Start and Finish times, and slack time, without a paywall.

For the Gantt chart, we chose TeamGantt to prioritize efficiency. Rather than manually drawing bars in a general diagramming application, TeamGantt's dedicated engine allows us to visualize start times, task durations, and resource allocation. This approach bridges the gap between high-level planning and precise execution while staying within the project's free software requirement.

**Task Time & Roadblock Table**

Tasks-

1. Initial Setup (Getting book number and other info)
    a. Unclear instructions
2. Going to the library- best case scenario: 3 mins, worst case: 10 mins (Aleksandra, Zoe)
    a. Falling (injury)
    b. Snow (bad weather)
3. Go to the MECC Printer
    a. The keycard doesn't work
    b. The printer is out of paper
    c. No more printing dollars
4. Finding the book-
    a. The book is lost/missing/displaced
    b. Forgotten call name
5. Travel back to MEC: 3 mins, worst case: 10 mins (Aleksandra, Zoe)
    a. Fall

6. Transcribing the book
    a. Backwards transcription is improperly understood
    b. The pages are missing
7. Proofread
    a. Major error found.
8. Printing the transcription
    a. Printer jam
9. Email Transcription
    a. Wifi Drops out
10. Hand Deliver to Professor
    a. Dr. S stepped out of the classroom

*The Expected Time calculation uses the formula: (Optimistic + (4 * Most Likely) + Pessimistic) / 6.*

| **Task** | **Setbacks** | **Resolution Options** | **Additional Time for Setback** | **Optimistic Time** | **Pessimistic Time** | **Most Likely Time** | **Expected Time** | |
|---|---|---|---|---|---|---|---|---|
| **A. Initial Setup** | | | | 2 min | 4 min | 3 min | 3 min | |
| | Unclear Instruction | Clarify with the Professor | +2 min | | | | | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| **B. Going to the Library** | | | | 3 mins | 18 | 4 min | 6 mins | |
| | Falling (injury) | If a health center is needed, alert the team. | +10 min | | | | | |
| | Snow (bad weather) | Take indoor paths. | +5 min | | | | | |
| **C. Finding the book** | | | | 3 | 21 | 4 | 7 min | |
| | The book is lost/missing/displaced | Try to find a library attendant | +15 | | | | | |
| | | Contact dr. S | +3 | | | | | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| **D. Finding the page** | | | | 3 min | 6 | 4 | 5 min | |
| | The pages are missing | Contact Dr. S | + 3 min | | | | | |
| **E. Travel back to the MECC** | | | | 5 | 18 | 4 | 7 min | |
| | Falling (injury) | If a health center is needed, alert the team. | +10 min | | | | | |
| | Snow (bad weather) | Take indoor paths. | +3 min | | | | | |
| **F. Transcribing the book** | | | | 10 | 27 | 15 | 16 min | |
| | Backwards transcription is improperly | Teammates will recover the paragraph | +7 mins | | | | | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | understood | | | | | | | |
| | The pages are missing | Contact dr S | +10 mins | | | | | |
| **G. Travel to MECC printer** | | | | 1 min | 19 mins | 4 mins | 6 mins | |
| | Falling | Send another team member and call for help | +5 | | | | | |
| | Unable to get in the door/No ID access | Try basement printer | +3 mins | | | | | |
| | | Try LSC Printer | +10 mins | | | | | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| **H. Email Transcription** | | | | 2 mins | 7 min | 4 min | 4 min | |
| | Wifi Drops out | Use phone hotspot | +1 min | | | | | |
| | MFA doesn’t send authentication token so can’t log into email | Another teammate will take over | +4 mins | | | | | |
| **I. Printing** | | | | +1 min | +3 min | 2 min | 2 min | |
| | Printer jam | Fix jam | +2 min | | | | | |
| **J. Proofreading** | | | | 3 mins | 6 mins | 4 mins | 4 min | |

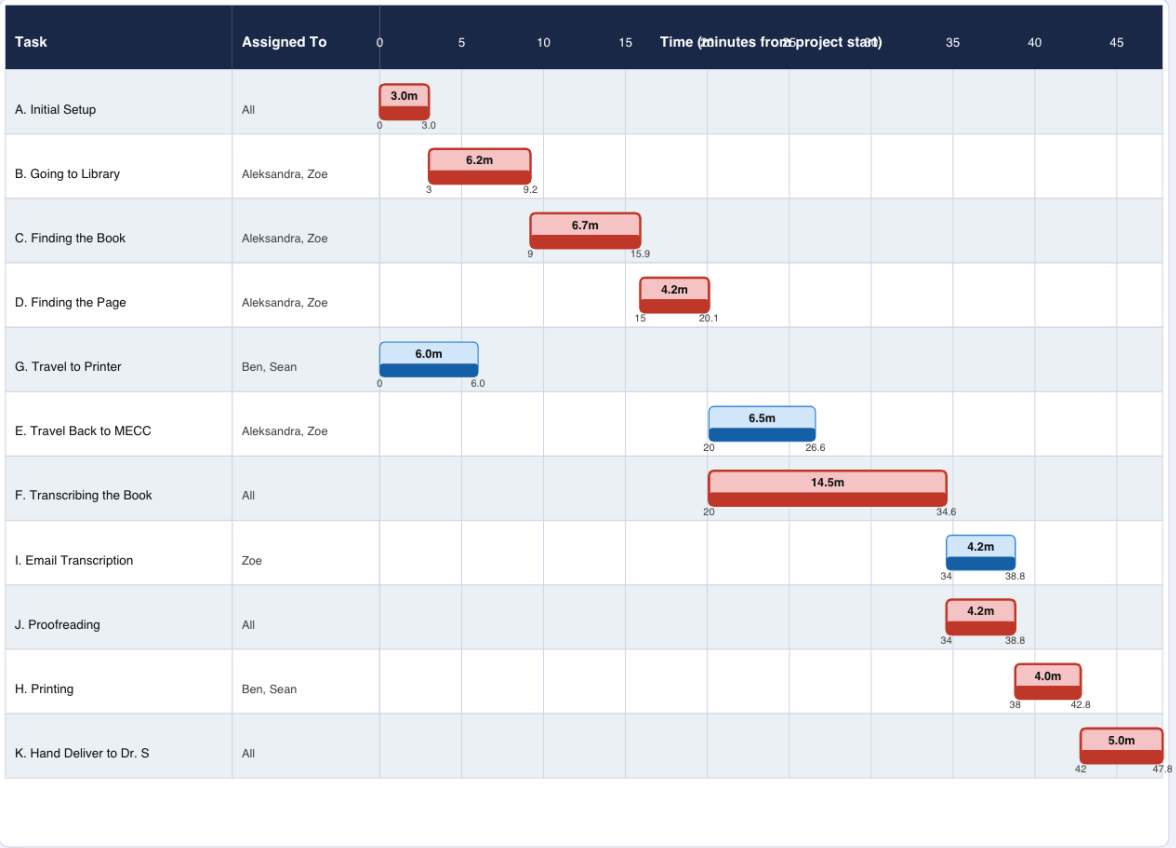
| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | Major error found. | Split up the paragraph into the # of teammates able to type and work together | +3 mins | | | | | |
| **K. Hand Deliver** | | | | 4 mins | 6 mins | 5 min | 5 mins | |
| | Dr. S stepped out of room | Check her office and send an email | +2 min | | | | | |

| Task Name | Expected Time | Predecessors |
|---|---|---|
| **A. Initial Setup** | 3 mins | None |
| **B. Going to the Library** | 6 mins | A |
| **C. Finding the book** | 7 mins | B |

| | | |
|---|---|---|
| **D. Finding the page** | 5 mins | C |
| **E. Travel back to the MECC** | 7 mins | D |
| **F. Transcribing the book** | 16 mins | D |
| **G. Travel to the MECC printer** | 6 min | None |
| **H. Printing** | 4 min | J |
| **I. Email Transcription** | 4 min | J |
| **J. Proofreading** | 4 mins | F |
| **K. Hand Deliver to Dr. S** | 5 mins | H,I |

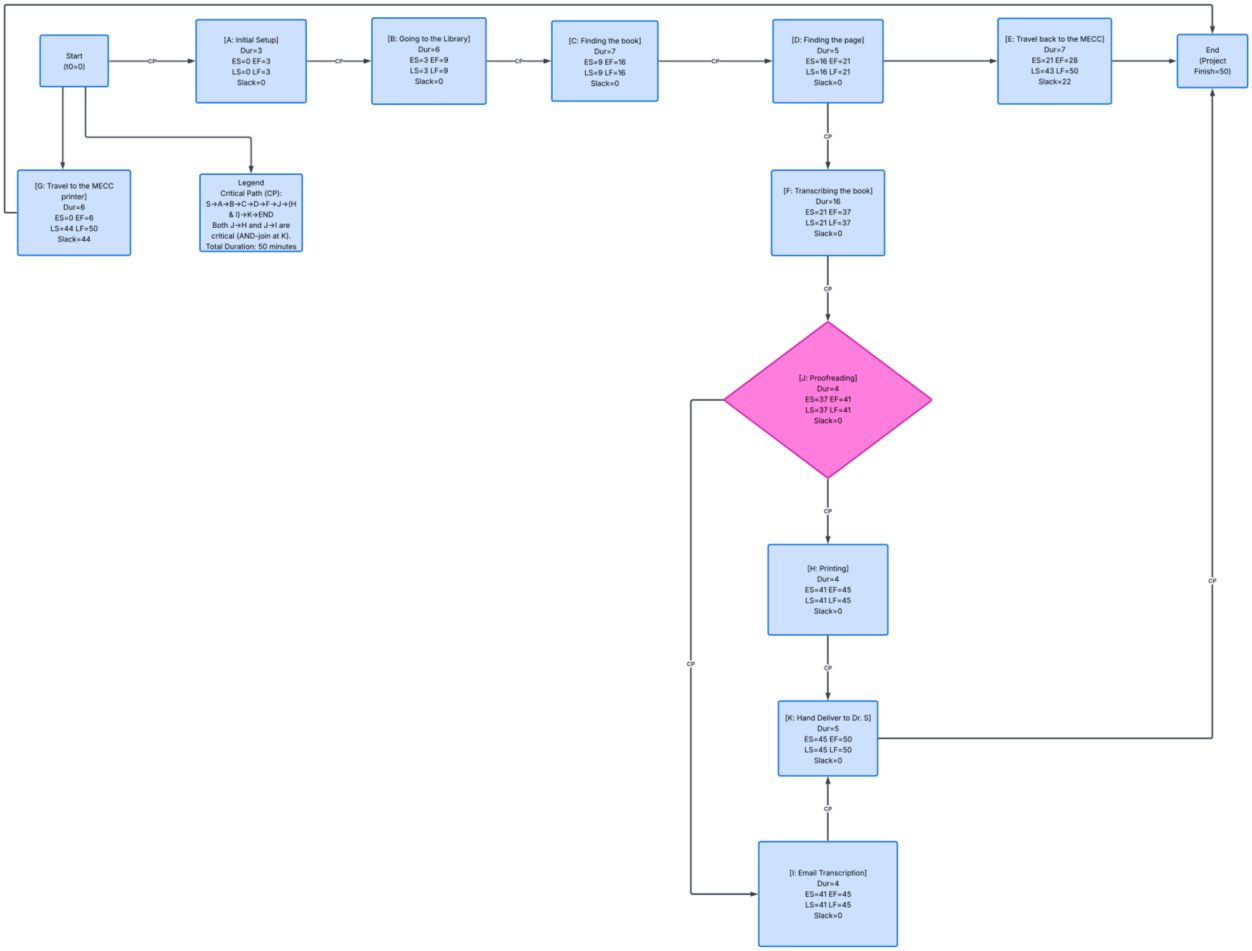
## Gantt Chart — Book Transcription Part 2

Project Duration: 47.8 min




Critical Path Task  Non-Critical Task  *Numbers below bars = start/end times (min). Critical path: A→B→C→D→F→J→H→K = 47.8 min*

Pert Chart:


Start
(t0=0)
[A: Initial Setup]
Dur=3
ES=0 EF=3
LS=0 LF=3
Slack=0
[B: Going to the Library]
Dur=6
ES=3 EF=9
LS=3 LF=9
Slack=0
[C: Finding the book]
Dur=7
ES=9 EF=16
LS=9 LF=16
Slack=0
[D: Finding the page]
Dur=5
ES=16 EF=21
LS=16 LF=21
Slack=0
[E: Travel back to the MECC]
Dur=7
ES=21 EF=28
LS=43 LF=50
Slack=22
End
(Project
Finish=50)
[G: Travel to the MECC
printer]
Dur=6
ES=0 EF=6
LS=44 LF=50
Slack=44
Legend
Critical Path (CP):
S→A→B→C→D→F→J→{H
& I}→K→END
Both J→H and J→I are
critical (AND-join at K).
Total Duration: 50 minutes
[F: Transcribing the book]
Dur=16
ES=21 EF=37
LS=21 LF=37
Slack=0
[J: Proofreading]
Dur=4
ES=37 EF=41
LS=37 LF=41
Slack=0
[H: Printing]
Dur=4
ES=41 EF=45
LS=41 LF=45
Slack=0
[K: Hand Deliver to Dr. S]
Dur=5
ES=45 EF=50
LS=45 LF=50
Slack=0
[I: Email Transcription]
Dur=4
ES=41 EF=45
LS=41 LF=45
Slack=0
CP